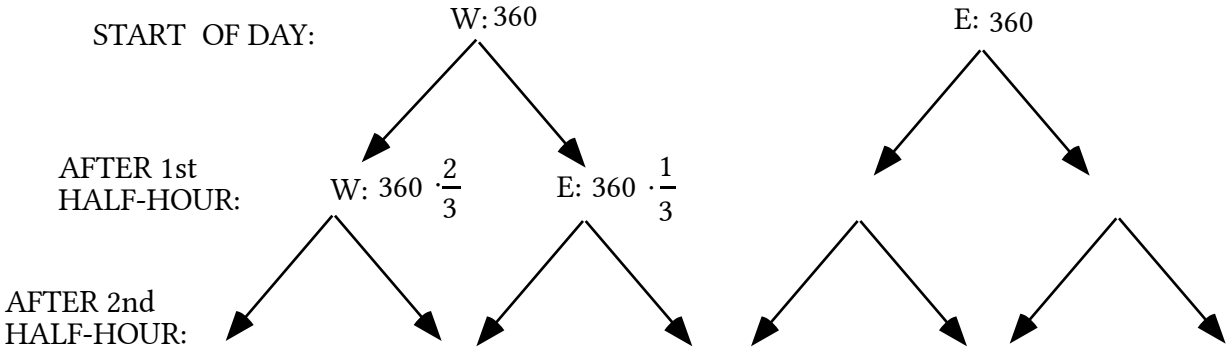
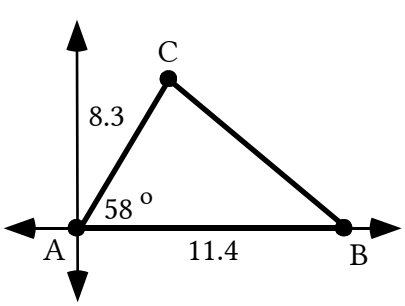
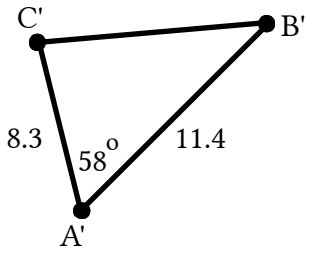
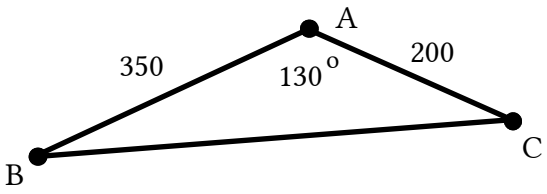
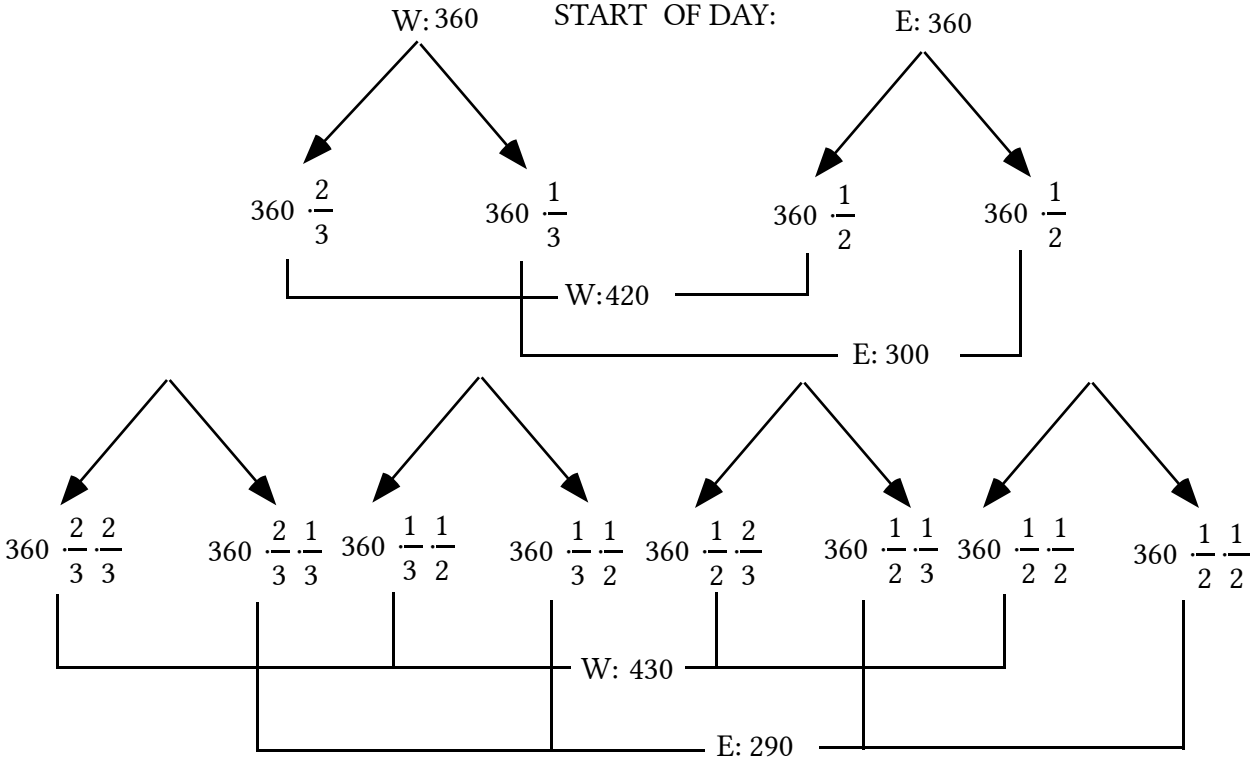

START OF DAY:
W: 360
E: 360
AFTER 1st HALF-HOUR:
W: 360 · 2/3
E: 360 · 1/3
AFTER 2nd HALF-HOUR:


C
8.3
58 o
A
11.4
B

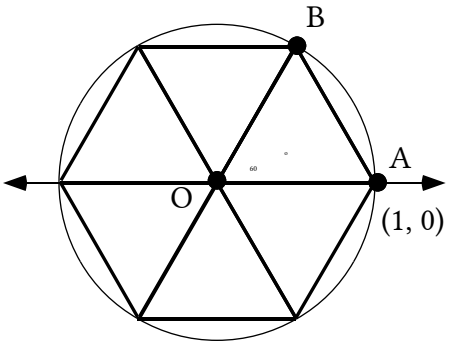

C'
B'
8.3
58 o
11.4
A'

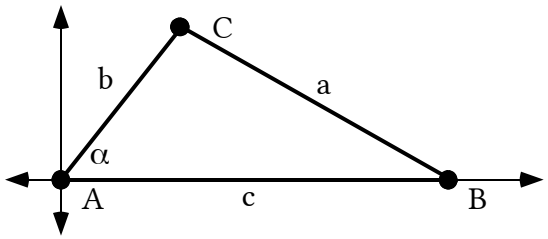

A
350
130 o
200
B
C

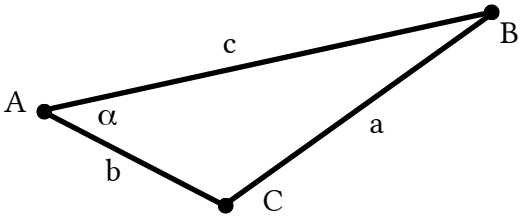

W: 360
START OF DAY:
E: 360
360 · 2/3
360 · 1/3
360 · 1/2
360 · 1/2
W: 420
E: 300
360 · 2/3 · 2/3
360 · 2/3 · 1/3
360 · 1/3 · 1/2
360 · 1/3 · 1/2
360 · 1/2 · 2/3
360 · 1/2 · 1/3
360 · 1/2 · 1/2
360 · 1/2 · 1/2
W: 430
E: 290

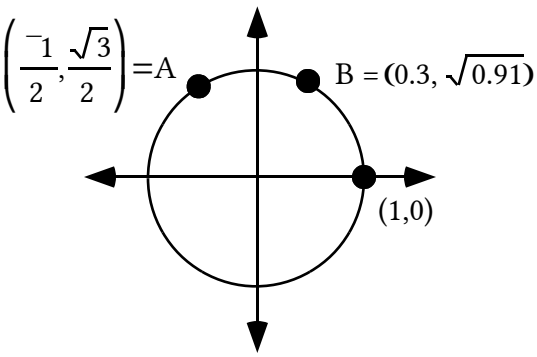
original disposition matrix | transition matrix | disposition after first half-hour | disposition after first half-hour

$$\begin{bmatrix} 360 & 360 \end{bmatrix} \cdot \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 360 \cdot \frac{2}{3} + 360 \cdot \frac{1}{2} & 360 \cdot \frac{1}{3} + 360 \cdot \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 420 & 300 \end{bmatrix}$$


B
A
O
(1, 0)

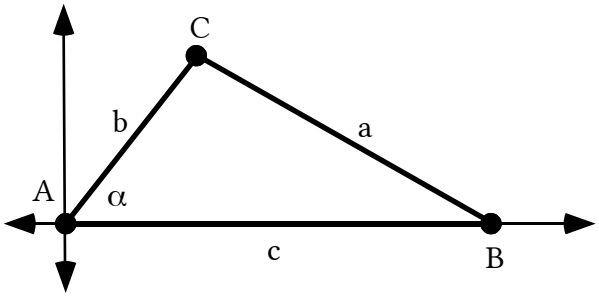

C
b
a
α
A
c
B

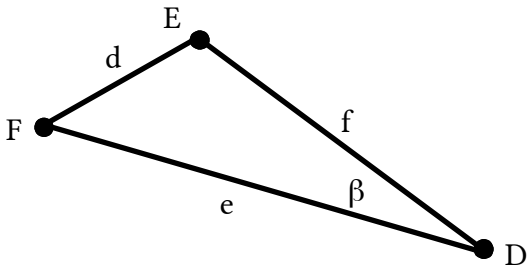

A
B
C
c
α
a
b

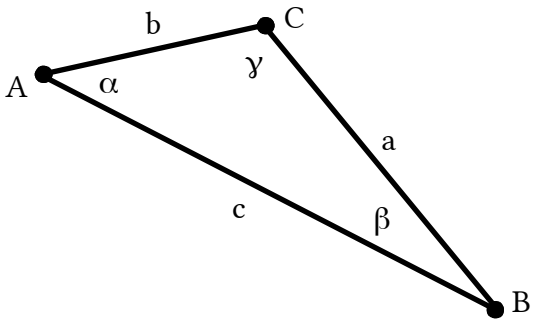

$\left(\frac{-1}{2}, \frac{\sqrt{3}}{2}\right)$=A
B = (0.3, $\sqrt{0.91}$)
(1,0)

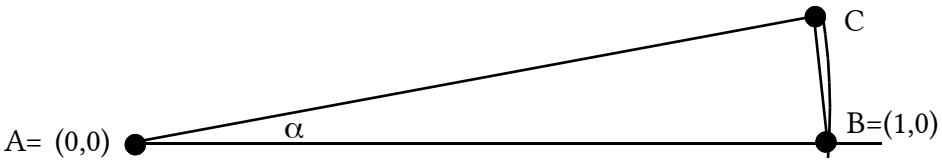
$$\begin{cases} \frac{-1}{3} \cdot x + \frac{1}{2} \cdot y = 0 \\ \frac{1}{3} \cdot x + \frac{-1}{2} \cdot y = 0 \\ x + y = 720 \end{cases}$$

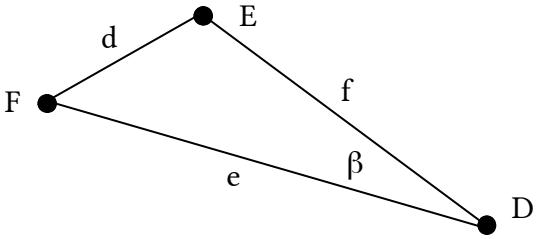
$$\begin{bmatrix} -\frac{1}{3} & \frac{1}{2} \\ \frac{1}{3} & -\frac{1}{2} \\ 1 & 1 \end{bmatrix}$$

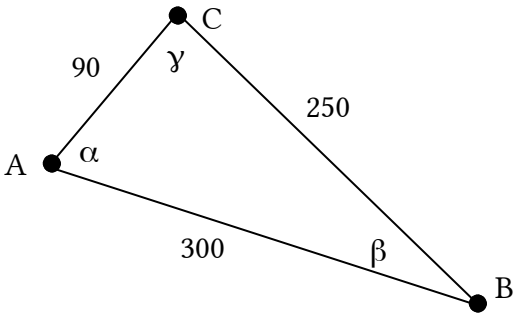

C
b
a
A
α
c
B

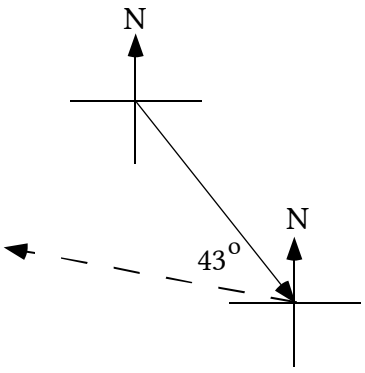

E
d
F
f
β
e
D


b
C
γ
A
α
a
c
β
B

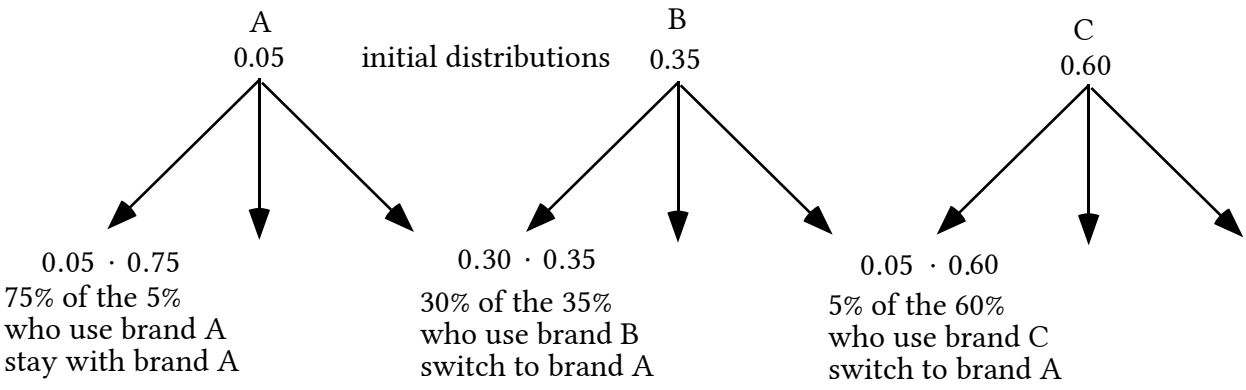

C
A= (0,0)
α
B=(1,0)

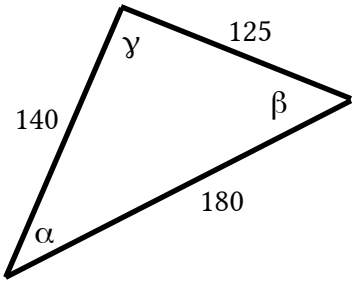

E
d
F
f
β
e
D

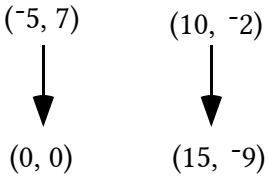

C
90
γ
250
A
α
300
β
B

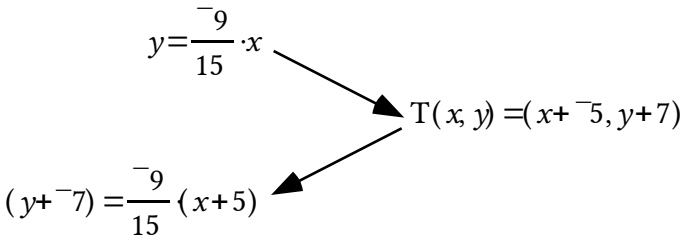

N
N
43°

| from \ to | brand A | brand B | brand C |
|---|---|---|---|
| brand A | 0.75 | 0.15 | 0.10 |
| brand B | 0.30 | 0.60 | 0.10 |
| brand C | 0.05 | 0.15 | 0.80 |

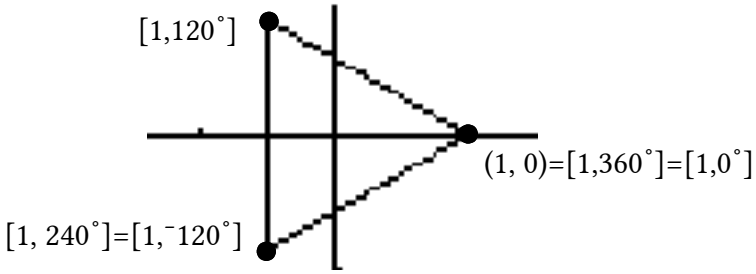

A
0.05
initial distributions
B
0.35
C
0.60
0.05 · 0.75
75% of the 5%
who use brand A
stay with brand A
0.30 · 0.35
30% of the 35%
who use brand B
switch to brand A
0.05 · 0.60
5% of the 60%
who use brand C
switch to brand A

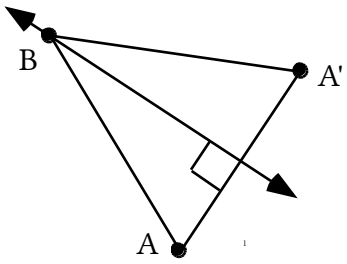

γ
125
β
140
180
α

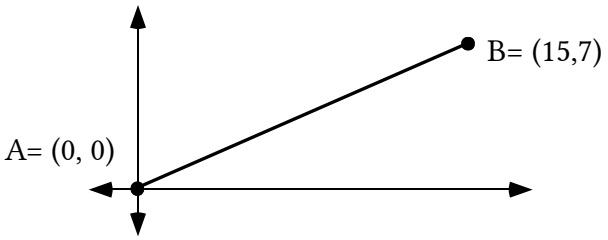

(-5, 7)
(10, -2)
(0, 0)
(15, -9)

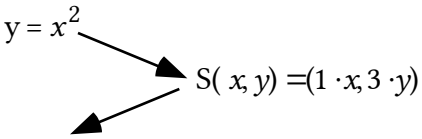

$y=\frac{-9}{15}\cdot x$
$T(x,y)=(x+-5,y+7)$
$(y+-7)=\frac{-9}{15}(x+5)$

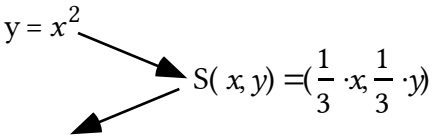
1 · 3

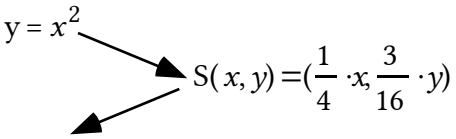

[1,120˚]
(1, 0)=[1,360˚]=[1,0˚]
[1, 240˚]=[1,-120˚]

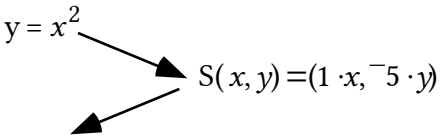

B
A'
A

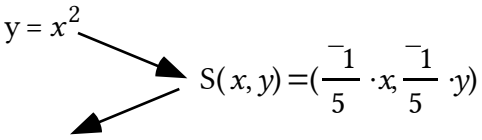

B= (15,7)
A= (0, 0)

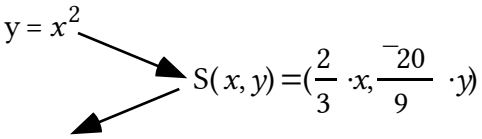

y = $x^2$
S( $x$, $y$) =(1 ·$x$, 3 ·$y$)

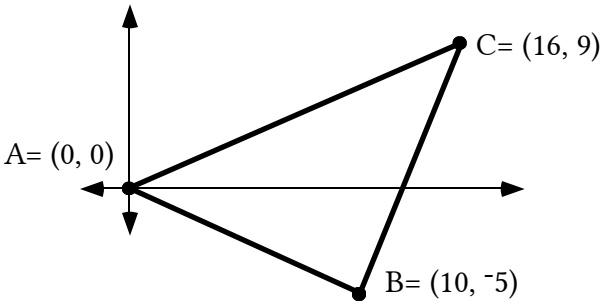

y = $x^2$
S( $x, y$) = ( $\frac{1}{3} \cdot x, \frac{1}{3} \cdot y$)

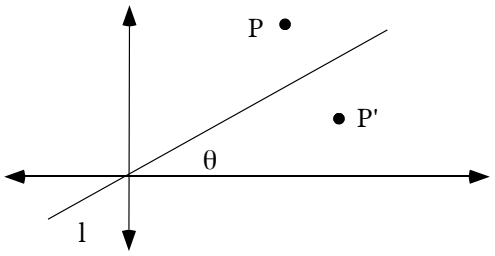

y = $x^2$
S( $x, y$) =( $\frac{1}{4}$ ·$x$, $\frac{3}{16}$ ·$y$)

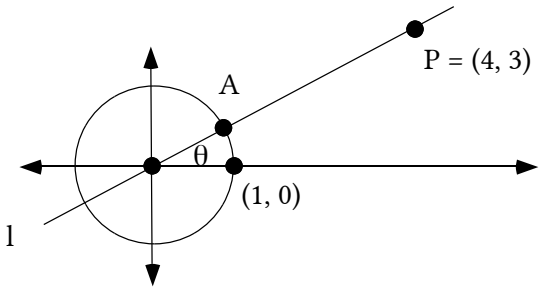

y = $x^2$
S( $x$, $y$) =(1 ·$x$, ¯5 ·$y$)

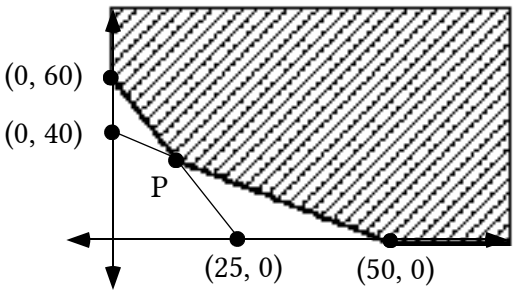

y = $x^2$
$S(x,y)=(\frac{1}{5}\cdot x, \frac{1}{5}\cdot y)$

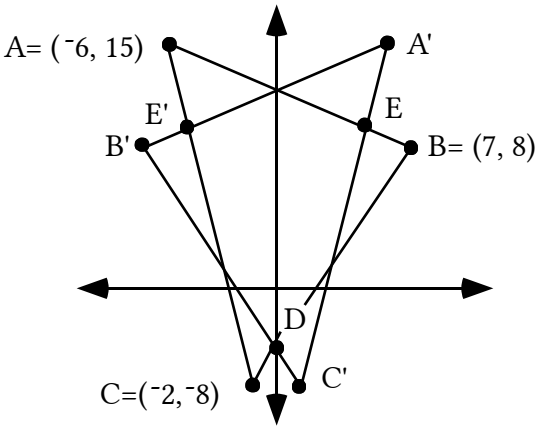

y = $x^2$
S( $x, y$) =( $\frac{2}{3}$ ·$x$, $\frac{\bar{}20}{9}$ ·$y$)

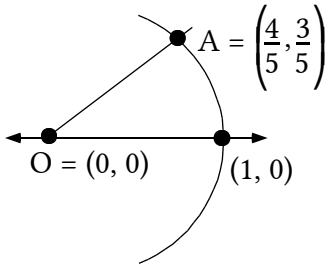

C= (16, 9)
A= (0, 0)
B= (10, -5)

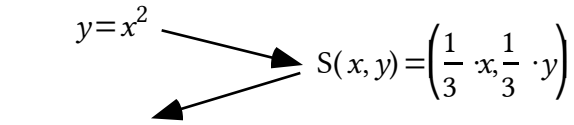

P
P'
θ
l


P = (4, 3)
A
θ
(1, 0)
l


(0, 60)
(0, 40)
P
(25, 0)
(50, 0)

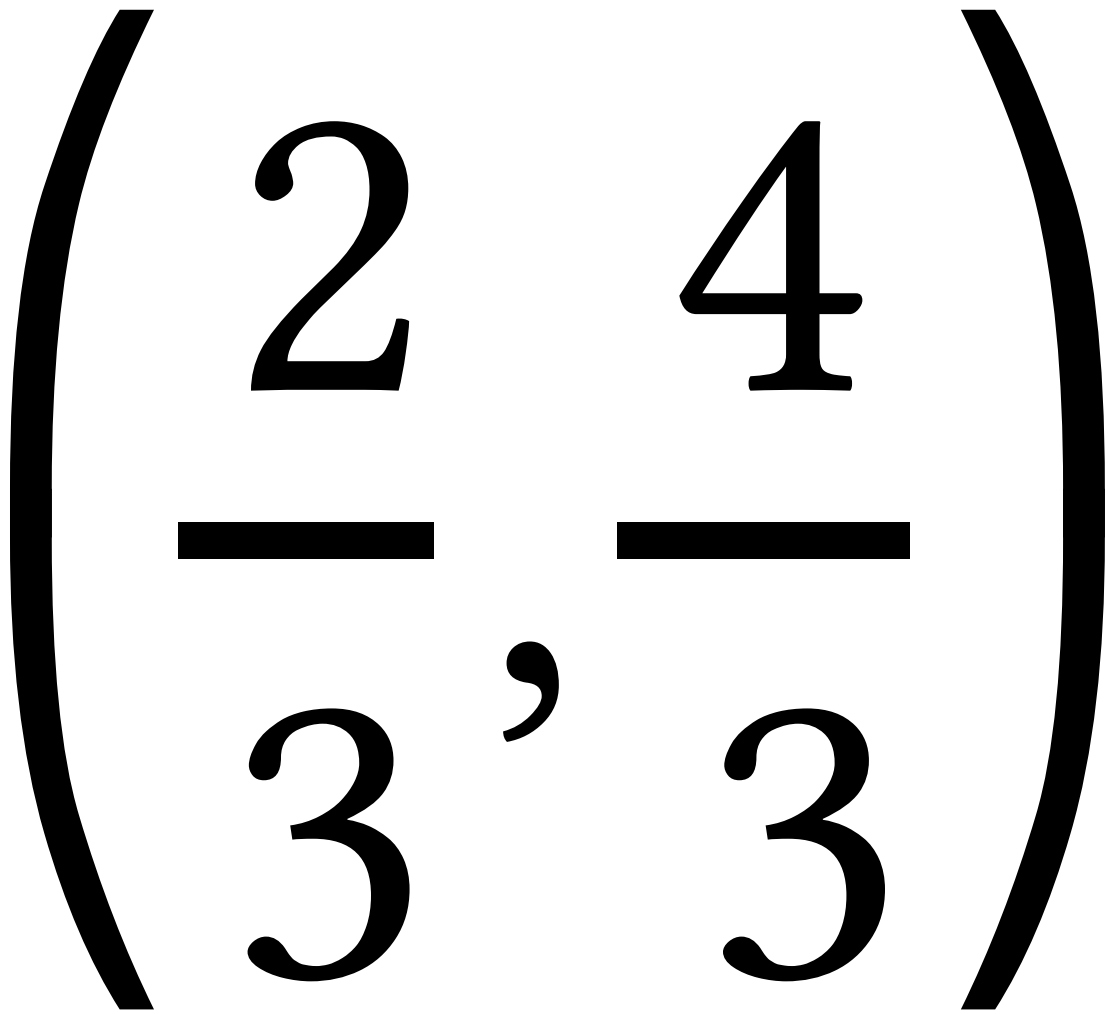

A= (⁻6, 15)
A'
E'
E
B'
B= (7, 8)
D
C=(⁻2,⁻8)
C'

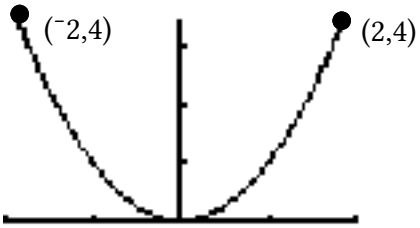

A = $\left(\frac{4}{5}, \frac{3}{5}\right)$
O = (0, 0)
(1, 0)



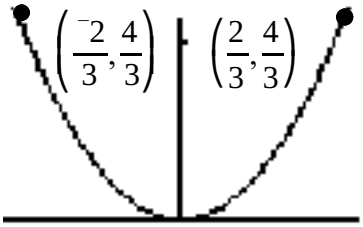

$(3 \cdot y) = (3 \cdot x)^2$ or $3 \cdot y = 9 \cdot x^2$ or $y = 3 \cdot x^2$

$$\left( \frac{-2}{3}, \frac{4}{3} \right)$$

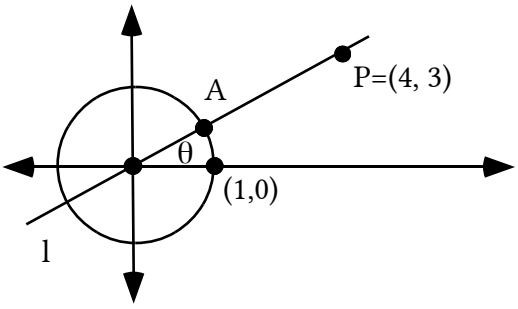


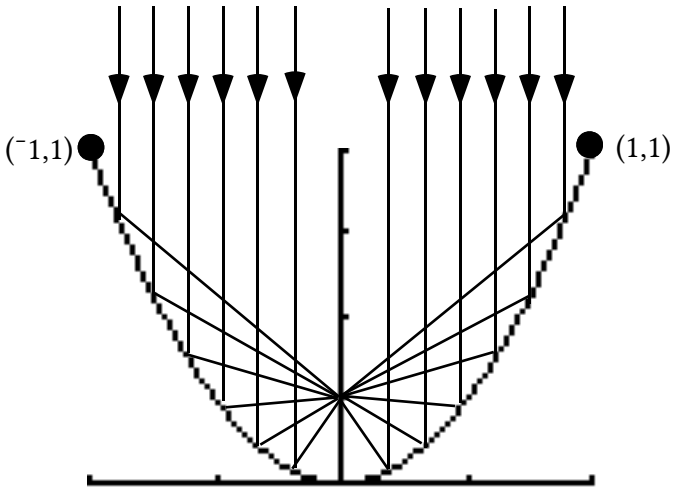



graph generated from $Y_1=X^2$ with Xmin = ⁻2, Xmax = 2, Ymin = 0 and Ymax = 4

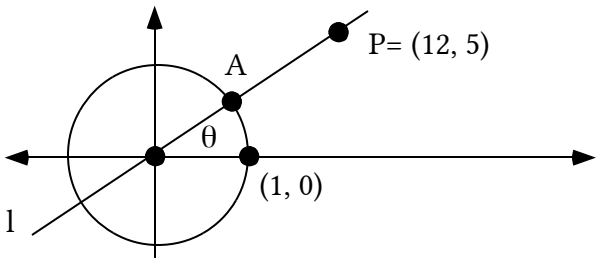


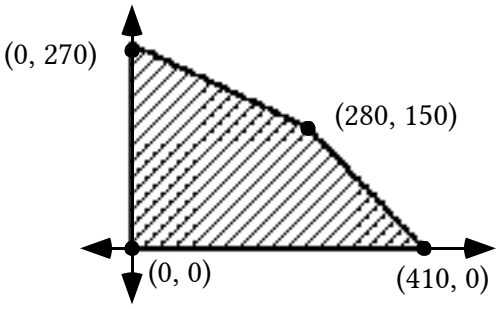

graph generated from $Y_2=3\cdot X^2$ with Xmin = $\frac{2}{3}$, Xmax = $\frac{-2}{3}$, Ymin = 0 and Ymax = $\frac{4}{3}$

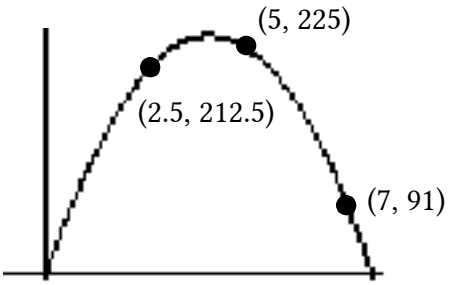

P=(4, 3)
A
θ
(1,0)
l

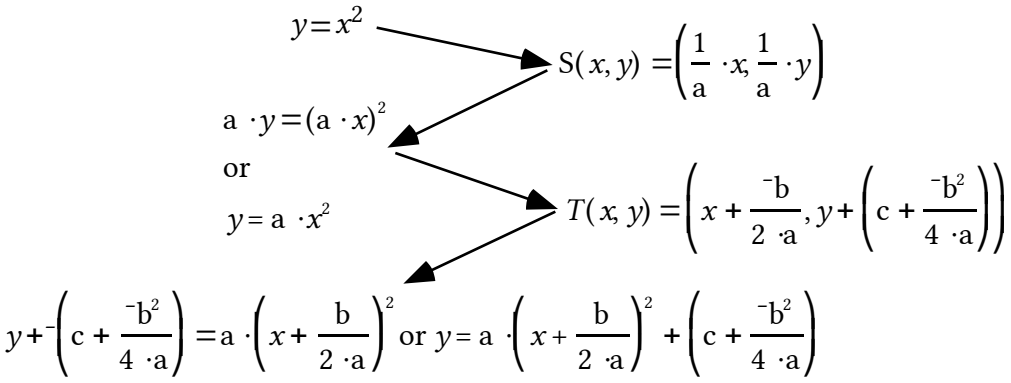


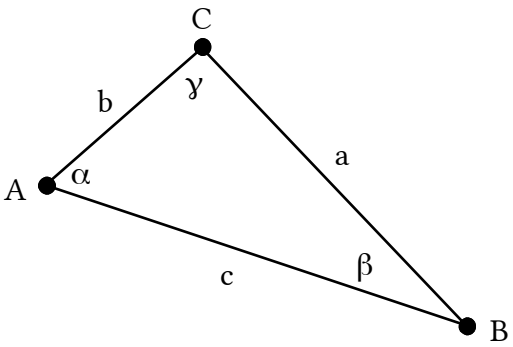

a representation of parallel light rays reflecting off a parabolic mirror to the focus.

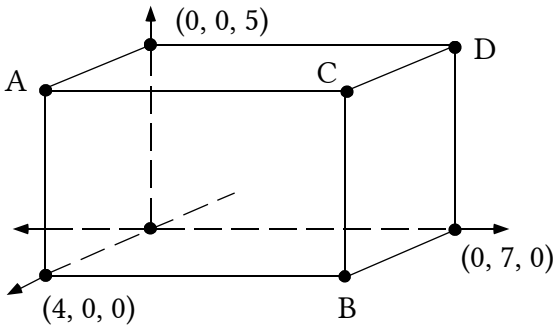

P= (12, 5)
A
θ
(1, 0)
l

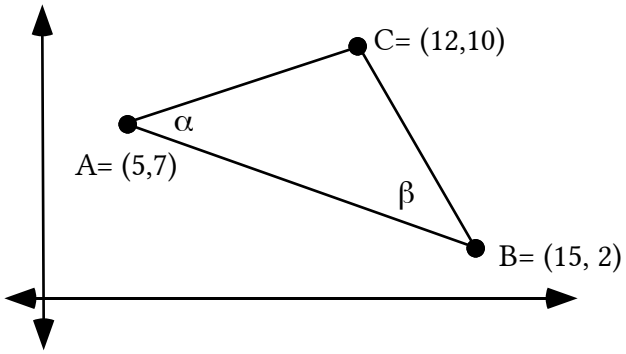

(0, 270)
(280, 150)
(0, 0)
(410, 0)

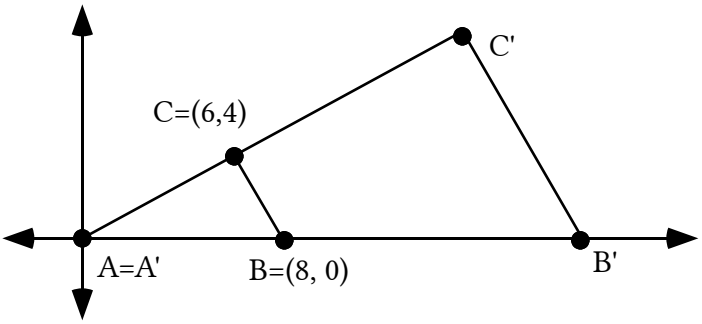

(5, 225)
(2.5, 212.5)
(7, 91)

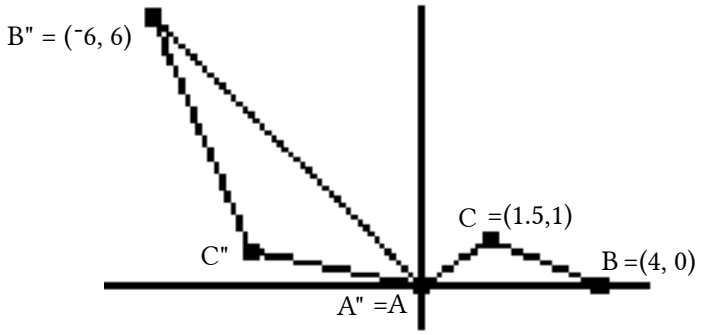

$y = x^2$
$S(x, y) = \left(\frac{1}{a} \cdot x, \frac{1}{a} \cdot y\right)$
$a \cdot y = (a \cdot x)^2$
or
$y = a \cdot x^2$
$T(x, y) = \left(x + \frac{-b}{2 \cdot a}, y + \left(c + \frac{-b^2}{4 \cdot a}\right)\right)$
$y + \left(-\left(c + \frac{-b^2}{4 \cdot a}\right)\right) = a \cdot \left(x + \frac{b}{2 \cdot a}\right)^2$ or $y = a \cdot \left(x + \frac{b}{2 \cdot a}\right)^2 + \left(c + \frac{-b^2}{4 \cdot a}\right)$

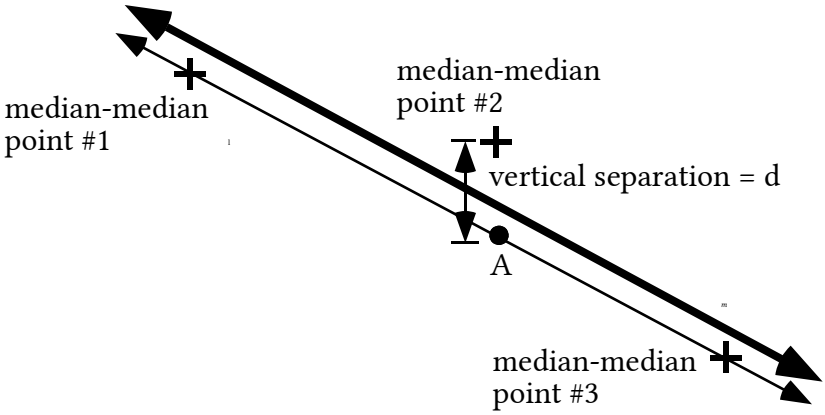

C
b
γ
a
A
α
β
c
B

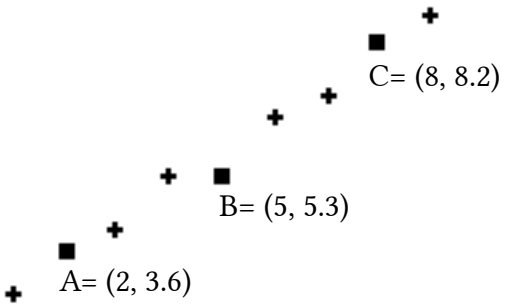

(0, 0, 5)
A
C
D
(0, 7, 0)
(4, 0, 0)
B

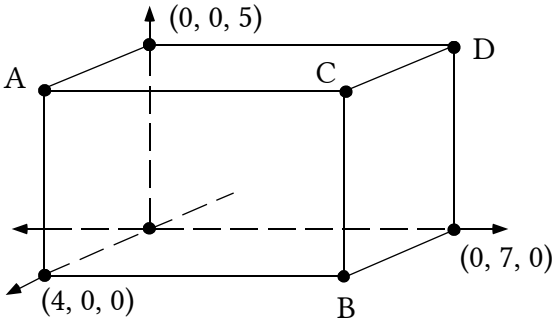

C= (12,10)
α
A= (5,7)
β
B= (15, 2)

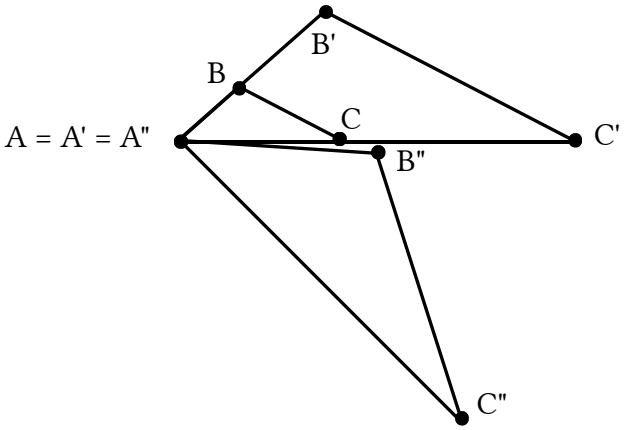

C'
C=(6,4)
A=A'
B=(8, 0)
B'


B" = (-6, 6)
C =(1.5,1)
C"
B =(4, 0)
A" =A

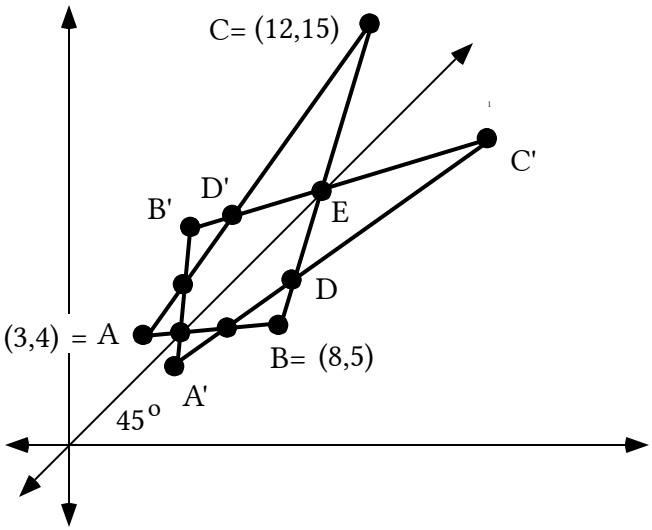

median-median
point #1
median-median
point #2
vertical separation = d
A
median-median
point #3

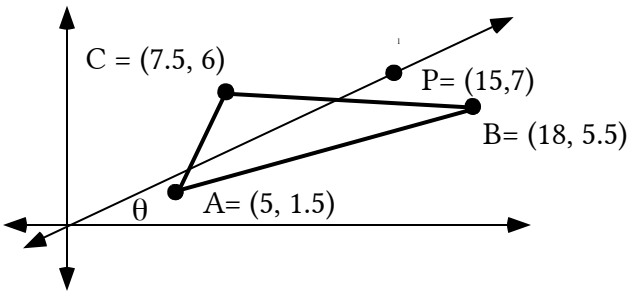

C= (8, 8.2)
B= (5, 5.3)
A= (2, 3.6)


(0, 0, 5)
A
C
D
(0, 7, 0)
(4, 0, 0)
B

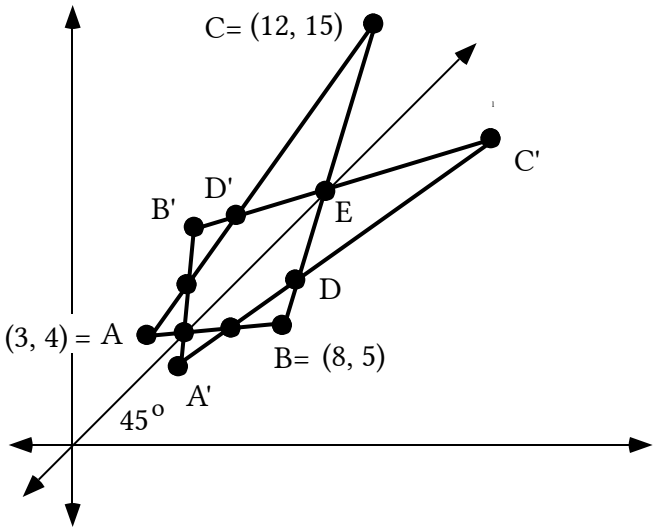

B'
B
C
A = A' = A"
C'
B"
C"

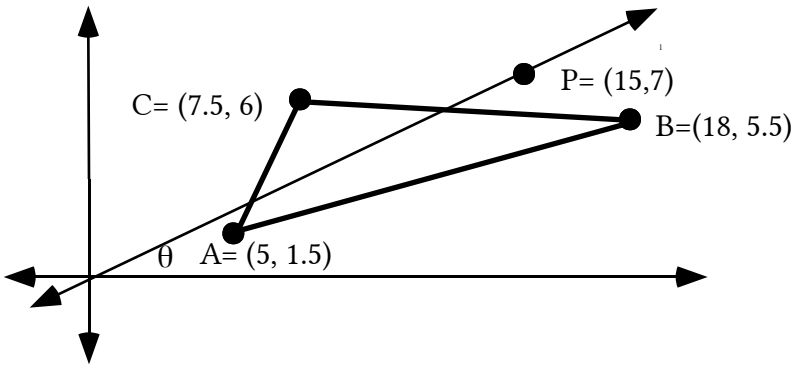
| time in seconds | distance in meters |
|---|---|
| 1 | 4.2 |
| 2 | 3.45 |
| 3 | 2.85 |
| 4 | 2.32 |
| 5 | 1.5 |
| 6 | 0.85 |


C= (12,15)
C'
D'
B'
E
D
(3,4) = A
B= (8,5)
A'
45°


C = (7.5, 6)
P= (15,7)
B= (18, 5.5)
θ
A= (5, 1.5)

| time in seconds | distance in meters |
|---|---|
| 1 | 1.95 |
| 2 | 3.17 |
| 3 | 4.31 |
| 4 | 5.22 |
| 5 | 6.4 |
| 6 | 7.64 |


C= (12, 15)
C'
D'
B'
E
D
(3, 4) = A
B= (8, 5)
A'
45°


P= (15,7)
C= (7.5, 6)
B=(18, 5.5)
θ
A= (5, 1.5)